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Inventor(s)

CHANG; Ting-Hsiang et al.

SEMICONDUCTOR SEED LAYER ON SOURCE/DRAIN DIELECTRIC STRUCTURES

Abstract

The present disclosure describes a semiconductor device having a semiconductor seed layer on a S/D dielectric structure. The semiconductor structure includes a channel structure on a substrate, a gate structure wrapped around the channel structure, an inner spacer adjacent to end portions of the gate structure and the channel structure, a dielectric structure on the substrate and adjacent to the inner spacer, a non-crystalline semiconductor material on the dielectric structure, and an epitaxial structure on top surfaces of the non-crystalline semiconductor material and the dielectric structure.

Inventors: CHANG; Ting-Hsiang (New Taipei, TW), KO; Chung-Ting (Kaohsiung City, TW), LIAO; Shu Ling (Taichung City, TW), LIN; Yu-Ru (Hsinchu, TW), LIN; Sung-En (Hsinchu County, TW), HUANG; Tai-Chun (Hsin-Chu City, TW), LEE; Tze-Liang (Hsinchu, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 96276947

Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/560,440, titled “Silicon Seed Layer on Embedded Dielectric for Bottom-Up Epitaxial Growth without DC Penalty,” filed Mar. 1, 2024, the disclosure of which is incorporated by reference in its entirety.

BACKGROUND

[0002] With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs and fin field effect transistors (finFETs). Such scaling down has increased the complexity of semiconductor manufacturing processes and increased the difficulty of process control in the semiconductor devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

[0004] FIG. 1 illustrates an isometric view of a semiconductor device having a semiconductor seed layer on source/drain dielectric structures, in accordance with some embodiments.

[0005] FIG. 2 illustrates a cross-sectional view of a semiconductor device having a semiconductor seed layer on source/drain dielectric structures, in accordance with some embodiments.

[0006] FIG. 3 is a flow diagram of a method for fabricating a semiconductor device having a semiconductor seed layer on source/drain dielectric structures, in accordance with some embodiments.

[0007] FIGS. 4-12 illustrate cross-sectional views of a semiconductor device having a semiconductor seed layer on source/drain dielectric structures at various stages of its fabrication, in accordance with some embodiments.

[0008] Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0009] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which

additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0010] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0011] It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

[0012] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0013] In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 20% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 10\%$, $\pm 20\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0014] With increasing demand for lower power consumption, higher performance, and smaller semiconductor devices, dimensions of semiconductor devices continue to scale down. The continuous scaling down of device dimensions and the increasing demand for device performance may require various process and material improvements, which can have multiple challenges. For example, nanostructure transistors can provide improved device performance with a channel in a stacked nanosheet/nanowire configuration. The nanostructure transistors can include finFETs, gate-all-around field effect transistors (GAA FETs), nanosheet transistors, nanowire transistors, multi-bridge channel transistors, nano-ribbon transistors, and other similar structured transistors. A source/drain (S/D) dielectric structure can be formed on the substrate between the stacked nanosheets/nanowires to reduce the leakage current through the substrate parasitic channel below the stacked nanosheets/nanowires. However, the subsequently grown S/D structures can include an amorphous portion on the S/D dielectric structure and a crystalline portion on the amorphous portion. The amorphous portion can have a higher resistance and a lower carrier mobility than the crystalline portion. Additionally, the amorphous portion can lead to a loss of strain on the channel of the nanostructure transistors. As a result, the direct current (DC) device performance of the nanostructure transistors may be reduced.

[0015] Various embodiments in the present disclosure provide methods for forming a semiconductor seed layer on S/D dielectric structures in a semiconductor device (e.g., a nanostructure transistor) and/or other semiconductor devices in an integrated circuit (IC). In some embodiments, a channel structure stacked on a fin structure can be formed on a substrate. A recess can be formed in the fin structure adjacent to the channel structure. A S/D dielectric structure and a

semiconductor seed layer can be formed in the recess. The S/D dielectric structure and the semiconductor seed layer can be below a bottom surface of the channel structure. An epitaxial structure can be formed on top surfaces of the S/D dielectric structure and the semiconductor seed layer. The epitaxial structure can include a non-crystalline portion and a crystalline portion. The crystalline portion can be in contact with the channel structure and the non-crystalline portion can be below the bottom surface of the channel structure. With the S/D dielectric structure, the semiconductor seed layer, and the non-crystalline portion of the epitaxial structure below the bottom surface of the channel structure, resistance between the epitaxial structure and the channel structure can be reduced. The carrier mobility in the epitaxial structure can be increased. The strain on the channel structure exerted by the epitaxial structure can be increased. Accordingly, the DC device performance of the semiconductor device can be improved.

[0016] FIG. 1 illustrates an isometric view of a semiconductor device **100** having a semiconductor seed layer on a S/D dielectric structure, in accordance with some embodiments. FIG. 2 illustrates a partial cross-sectional view of semiconductor device **100** having a semiconductor seed layer on S/D dielectric structures across line A-A shown in FIG. 1, in accordance with some embodiments.

[0017] In some embodiments, semiconductor device **100** can include transistors **102A-102C**, as shown in FIG. 1. In some embodiments, transistors **102A-102C** can include nanostructure transistors. The nanostructure transistors can include finFETs, GAA FETs, nanosheet transistors, nanowire transistors, multi-bridge channel transistors, nano-ribbon transistors, and other similar structured transistors. The nanostructure transistors can provide a channel in a stacked nanosheet/nanowire configuration. In some embodiments, transistors **102A-102C** can be n-type field-effect transistors (NFETs). In some embodiments, transistors **102A-102C** can be p-type field-effect transistors (PFETs). In some embodiments, any of transistors **102A-102C** can be an NFET or a PFET. Though FIG. 1 shows three transistors, semiconductor device **100** can have any number of transistors. In addition, semiconductor device **100** can be incorporated into an IC through the use of other structural components, such as conductive vias, conductive lines, dielectric layers, passivation layers, and interconnects, which are not shown for simplicity. The discussion of elements of transistors **102A-102C** with the same annotations applies to each other, unless mentioned otherwise. And like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

[0018] Referring to FIGS. 1 and 2, semiconductor device **100** having transistors **102A-102C** can be formed on a substrate **104** and can be isolated by shallow trench isolation (STI) regions **106**. Each of transistors **102A-102C** can include fin structures **108**, sidewall spacers **109**, gate dielectric layer **124**, gate structures **112**, gate spacers **114**, inner spacers **121**, S/D dielectric structures **111**, semiconductor seed layer **113**, S/D structures **110**, etch stop layer (ESL) **116**, interlayer dielectric (ILD) layer **118**, and S/D contact structures **130**. In some embodiments, as shown in FIG. 2, transistors **102A-102C** can have nanostructures **122-1**, **122-2**, and **122-3** (collectively referred to as “nanostructures **122**”) on fin structures **108**.

[0019] Referring to FIGS. 1 and 2, substrate **104** can include a semiconductor material, such as silicon. In some embodiments, substrate **104** includes a crystalline silicon substrate (e.g., silicon wafer). In some embodiments, substrate **104** includes (i) an elementary semiconductor, such as germanium; (ii) a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; (iii) an alloy semiconductor including silicon germanium carbide, silicon germanium, gallium arsenic phosphide, and/or aluminum gallium arsenide; or (iv) a combination thereof. Further, substrate **104** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **104** can be doped with p-type dopants (e.g., boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic).

[0020] STI regions **106** can provide electrical isolation between transistors **102A-102C** and from neighboring transistors (not shown) on substrate **104** and/or neighboring active and passive

elements (not shown) integrated with or deposited on substrate **104**. STI regions **106** can be made of a dielectric material. In some embodiments, STI regions **106** can include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating materials. In some embodiments, STI regions **106** can include a multi-layered structure.

[0021] Referring to FIGS. **1** and **2**, nanostructures **122** and fin structures **108** can be formed on patterned portions of substrate **104**. Embodiments of the nanostructures and fin structures disclosed herein may be patterned by any suitable method. For example, the nanostructures and fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can combine photolithography and self-aligned processes, forming patterns that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers can be formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers can then be used to pattern the nanostructures and fin structures.

[0022] As shown in FIGS. **1** and **2**, nanostructures **122** and fin structures **108** can extend along an X-axis for transistors **102A-102C**. In some embodiments, nanostructures **122** and fin structures **108** can be disposed on substrate **104**. Nanostructures **122** can include a stack of nanostructures **122-1**, **122-2**, and **122-3**, which can be in the form of nanosheets, nanowires, or nano-ribbons. Each of nanostructures **122** can act as a channel structure and form a channel region underlying gate structures **112** of transistors **102A-102C**. In some embodiments, nanostructures **122** and fin structures **108** can include semiconductor materials similar to or different from substrate **104**. In some embodiments, nanostructures **122** and fin structures **108** can include silicon. In some embodiments, nanostructures **122** and fin structures **108** can include silicon germanium. The semiconductor materials of nanostructures **122** and fin structures **108** can be undoped or can be in-situ doped during their formation process. In some embodiments, as shown in FIG. **2**, nanostructures **122** under gate structures **112** can form channel regions of semiconductor device **100** and represent current carrying channel structures of semiconductor device **100**. Though three layers of nanostructures **122** are shown in FIG. **2**, transistors **102A-102C** can have any number of nanostructures **122**. In some embodiments, nanostructures **122** can have a thickness $122t$ along a Z-axis ranging from about 3 nm to about 8 nm. In some embodiments, a spacing between adjacent nanostructures **122** along a Z-axis can range from about 5 nm to about 12 nm.

[0023] Referring to FIGS. **1** and **2**, gate dielectric layer **124** can be formed on nanostructures **122**, fin structures **108**, and STI regions **106**. In some embodiments, gate dielectric layer **124** can be multi-layered structures and can include an interfacial layer and a high-k dielectric layer. In some embodiments, gate dielectric layer **124** can include no interfacial layer and the high-k dielectric layer can be in direct contact with nanostructures **122**. In some embodiments, the interfacial layer can include silicon oxide formed by a deposition process or an oxidation process. In some embodiments, the interfacial layer can have a thickness ranging from about 0.1 nm to about 1.5 nm. In some embodiments, the high-k dielectric layer can include hafnium oxide, zirconium oxide, or other suitable high-k dielectric materials.

[0024] In some embodiments, as shown in FIGS. **1** and **2**, gate structures **112** can be disposed on gate dielectric layer **124**. In some embodiments, gate structures **112** can include one or more work function metal layers and a metal fill. The one or more work function metal layers can include work function metals to tune the threshold voltage ($V_{sub.t}$) of transistors **102A-102C**. In some embodiments, gate structures **112** for NFET and PFET devices can have substantially the same work-function metal. In some embodiments, gate structures **112** for NFET and PFET devices can have different work-function metals. In some embodiments, as shown in FIG. **2**, each of nanostructures **122** can be wrapped around by gate structures **112**, for which gate structures **112** can

be referred to as “gate-all-around (GAA) structures” and transistors **102A-102C** can also be referred to as “GAA FETs **102A-102C**.” The one or more work function metal layers can wrap around nanostructures **122** and can include work function metals to tune the $V_{sub.t}$ of transistors **102A-102C**. In some embodiments, transistors **102A-102C** can include any number of work function metal layers for $V_{sub.t}$ tuning (e.g., ultra-low $V_{sub.t}$, low $V_{sub.t}$, and standard $V_{sub.t}$). [0025] In some embodiments, NFETs **102A-102C** can include n-type work function metal layers. The n-type work function metal layers can include aluminum, titanium aluminum, titanium aluminum carbon, tantalum aluminum, tantalum aluminum carbon, tantalum silicon carbide, hafnium carbide, silicon, titanium nitride, titanium silicon nitride, or other suitable work function metals. In some embodiments, PFETs **102A-102C** can include p-type work function metal layers. The p-type work function metal layers can include titanium nitride, titanium silicon nitride, tantalum nitride, tungsten carbon nitride, tungsten, molybdenum, or other suitable work function metals. In some embodiments, the work function metal layers can include a single metal layer or a stack of metal layers. The stack of metal layers can include work function metals having work-function values equal to or different from each other. In some embodiments, the metal fill can include titanium, tantalum, aluminum, cobalt, tungsten, nickel, ruthenium, or other suitable conductive materials.

[0026] Referring to FIGS. **1** and **2**, gate spacers **114** can be disposed on sidewalls of gate structures **112** and in contact with gate dielectric layer **124**, according to some embodiments. Sidewall spacers **109** can be disposed on sidewalls of fin structures **108**. Inner spacers **121** can be disposed adjacent to end portions of nanostructures **122** and between S/D structures **110** and gate structures **112**. Gate spacers **114**, sidewall spacers **109**, and inner spacers **121** can include insulating materials, such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon oxycarbide, silicon oxycarbonitride, aluminum oxide, a low-k material, and a combination thereof. In some embodiments, gate spacers **114**, sidewall spacers **109**, and inner spacers **121** can include the same insulating material. In some embodiments, gate spacers **114**, sidewall spacers **109**, and inner spacers **121** can include different insulating materials. In some embodiments, gate spacers **114**, sidewall spacers **109**, and inner spacers **121** can include a single layer or a stack of insulating layers. In some embodiments, gate spacers **114**, sidewall spacers **109**, and inner spacers **121** can have a low-k material with a dielectric constant less than about 3.9 (e.g., about 3.5, about 3.0, or about 2.8).

[0027] In some embodiments, S/D dielectric structures **111** can be disposed on fin structures **108**. In some embodiments, S/D dielectric structures **111** can include aluminum oxide (AlO_{sub.x}), silicon carbide (SiC_{sub.x}), silicon nitride (SiN_{sub.x}), silicon carbonitride (SiC_{sub.x}N_{sub.1-x}), silicon oxycarbonitride (SiO_{sub.y}C_{sub.x}N_{sub.1-x-y}), a low-k material, or a combination thereof. In some embodiments, S/D dielectric structures **111** can reduce leakage current through fin structures **108** and improve device performance. In some embodiments, S/D dielectric structures **111** can have a thickness **111t** ranging from about 3 nm to about 5 nm. A ratio of thickness **111t** to thickness **122t** can range from about 0.2 to about 0.8. If thickness **111t** is less than about 3 nm or the ratio is less than about 0.2, S/D dielectric structures **111** may not reduce leakage current in semiconductor device **100** and device performance may not be improved. If thickness **111t** is greater than about 5 nm or the ratio is greater than about 0.8, S/D dielectric structures **111** may be in contact with nanostructures **122**, which can lead to reduced device current and degraded device performance.

[0028] In some embodiments, as shown in FIGS. **1** and **2**, S/D dielectric structures **111** can extend into fin structures **108**. In some embodiments, S/D dielectric structures **111** can extend a distance **111d** below a bottom surface of gate structures **112**. In some embodiments, distance **111d** can range from about 4 nm to about 12 nm. A ratio between distance **111d** to thickness **122t** can range from about 1 to about 2. If distance **111d** is less than about 4 nm or the ratio is less than about 1, device current between nanostructures **122** and S/D structures **110** can be reduced and device performance

can be degraded. If distance **111d** is greater than about 12 nm, or the ratio is greater than about 2, device performance may not be further improved while manufacturing cost may increase.

[0029] In some embodiments, S/D dielectric structures **111** can be below a bottom surface of bottom nanostructures **122-1** to avoid contact between S/D dielectric structures **111** and nanostructures **122**. If S/D dielectric structures **111** are in contact with nanostructures **122**, device current between nanostructures **122** and S/D structures **110** can be reduced and device performance can be degraded. In some embodiments, as shown in FIG. 2, S/D dielectric structures **111** can have a “U” shape and end portions of S/D dielectric structures **111** can cover inner spacers **121** below bottom nanostructures **122-1**. In some embodiments, as shown in FIG. 2, the end portions of S/D dielectric structures **111** can have sloped top surfaces with inner edges of the end portions lower than the outer edges of the end portions. In some embodiments, a height difference between the inner edges and the outer edges of the end portions of S/D dielectric structures **111** can range from about 5 nm to about 12 nm. In some embodiments, with end portions of S/D dielectric structures **111** covering inner spacers **121**, the parasitic capacitance between S/D structures **110** and gate structures **112** can be reduced and device performance can be improved.

[0030] In some embodiments, as shown in FIG. 2, semiconductor seed layer **113** can be disposed on S/D dielectric structures **111** and surrounded by S/D dielectric structures **111**. In some embodiments, semiconductor seed layer **113** can include non-crystalline silicon, non-crystalline silicon germanium, or other suitable non-crystalline semiconductor materials. In some embodiments, semiconductor seed layer **113** can include a non-crystalline semiconductor material, such as amorphous silicon or poly-crystalline silicon. In some embodiments, semiconductor seed layer **113** can act as a seed layer for subsequent growth of S/D structures **110**. Semiconductor seed layer **113** can facilitate the growth of S/D structures **110** without forming voids between S/D structures **110** and S/D dielectric structures **111**. In some embodiments, semiconductor seed layer **113** can be below the bottom surface of bottom nanostructures **122-1**. If semiconductor seed layer **113** is above the bottom surface of bottom nanostructures **122-1**, nanostructures **122** may be in contact with S/D dielectric structures **111** and/or semiconductor seed layer **113**, device current between nanostructures **122** and S/D structures **110** can be reduced and device performance can be degraded.

[0031] In some embodiments, semiconductor seed layer **113** can have a thickness **113t** ranging from about 1 nm to about 7.5 nm. A ratio of thickness **113t** to thickness **122t** can range from about 0.1 to about 0.9. If thickness **113t** is less than about 1 nm or the ratio is less than about 0.1, voids may be formed between S/D dielectric structures **111** and S/D structures **110** and the device performance may be degraded. If thickness **113t** is greater than about 7.5 nm or the ratio is greater about 0.9, nanostructures **122** may be in contact with non-crystalline semiconductor materials in semiconductor seed layer **113** and/or S/D structures **110** and the device current may be reduced.

[0032] Referring to FIGS. 1 and 2, S/D structures **110** can be disposed on top surfaces of semiconductor seed layer **113** and S/D dielectric structures **111**. S/D structures **110** can function as S/D regions of transistors **102A-102C**. In some embodiments, S/D structures **110** can have any geometric shape, such as a polygon, a cone, a diamond, an ellipsis, and a circle. In some embodiments, S/D structures **110** can include an epitaxially-grown semiconductor material, such as silicon (e.g., the same material as substrate **104**). In some embodiments, S/D structures **110** can include an epitaxially-grown semiconductor material different from the material of substrate **104**, such as silicon germanium, and can impart a strain on the channel regions of nanostructures **122** under gate structures **112**. Since the lattice constant of such epitaxially-grown semiconductor material is different from the material of substrate **104**, the channel regions of nanostructures **122** are strained to increase carrier mobility in the channel regions of semiconductor device **100**. The epitaxially-grown semiconductor material can include: (i) a semiconductor material, such as germanium and silicon; (ii) a compound semiconductor material, such as gallium arsenide and aluminum gallium arsenide; or (iii) a semiconductor alloy, such as silicon germanium and gallium

arsenide phosphide.

[0033] In some embodiments, S/D structures **110** can include silicon and can be in-situ doped during an epitaxial growth process using n-type dopants, such as phosphorus and arsenic. For n-type in-situ doping, n-type doping precursors, such as phosphine, arsine, and other n-type doping precursors, can be used. In some embodiments, S/D structures **110** can include silicon, silicon germanium, germanium, or III-V materials (e.g., indium antimonide, gallium antimonide, or indium gallium antimonide) and can be in-situ doped during an epitaxial growth process using p-type dopants, such as boron, indium, and gallium. For p-type in-situ doping, p-type doping precursors, such as diborane (B.sub.2H.sub.6), boron trifluoride (BF.sub.3), and other p-type doping precursors, can be used.

[0034] In some embodiments, as shown in FIG. 2, S/D structures **110** can have a height **110h** along a Z-axis ranging from about 50 nm to about 80 nm. In some embodiments, S/D structures **110** can have a width **110w** along an X-axis ranging from about 20 nm to about 40 nm. In some embodiments, as shown in FIG. 2, S/D structures **110** can include a non-crystalline portion **110A** and a crystalline portion **110B**. In some embodiments, non-crystalline portion **110A** can include a non-crystalline semiconductor material, such as amorphous or poly-crystalline semiconductor material. In some embodiments, non-crystalline portion **110A** can be below the bottom surface of bottom nanostructures **122-1**. If non-crystalline portion **110A** is above the bottom surface of bottom nanostructures **122-1**, the size of crystalline portion **110B** may be reduced, the strain imparted on the channel regions of nanostructures **122** may be reduced, and the carrier mobility in the channel regions of nanostructures **122** may be decreased. Additionally, if non-crystalline portion **110A** is above the bottom surface of bottom nanostructures **122-1**, nanostructures **122** may be in contact with non-crystalline portion **110A**, device current between nanostructures **122** and S/D structures **110** can be reduced and device performance can be degraded.

[0035] In some embodiments, non-crystalline portion **110A** can have a thickness **110At** along a Z-axis ranging from about 1 nm to about 6 nm. In some embodiments, a ratio between thickness **110At** and thickness **122t** can range from 0.3 to about 0.7. If thickness **110At** is less than about 1 nm or the ratio is less than about 0.3, crystalline portion **110B** of S/D structures may become non-crystalline. If thickness **110At** is greater than about 6 nm or the ratio is greater than about 0.7, the size of crystalline portion **110B** may be reduced, the strain on nanostructures **122** and the carrier mobility in nanostructures **122** may be decreased, non-crystalline portion **110A** may be in contact with nanostructures **122**, drive current of semiconductor device **100** may be reduced, and device performance of semiconductor device **100** may be degraded.

[0036] In some embodiments, crystalline portion **110B** of S/D structures **110** can include one or more epitaxial layers, where each epitaxial layer can have different compositions. In some embodiments, each of the one or more epitaxial layers can include Si and differ from each other based on, for example, doping concentration and/or epitaxial growth process conditions. In some embodiments, each of the one or more epitaxial layers can include silicon germanium and differ from each other based on, for example, doping concentration, epitaxial growth process conditions, and/or relative concentration of germanium with respect to silicon.

[0037] Referring to FIGS. 1 and 2, ESL **116** can be disposed on S/D structures **110** and sidewalls of gate spacers **114** and sidewall spacers **109**. ESL **116** can be configured to protect S/D structures **110**, S/D dielectric structures **111**, and gate structures **112** during subsequent formation of S/D contact structures **130** on S/D structures **110**. In some embodiments, ESL **116** can include, for example, silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon carbonitride, boron nitride, silicon boron nitride, silicon carbon boron nitride, or a combination thereof.

[0038] ILD layer **118** can be disposed on ESL **116** over S/D structures **110** and STI regions **106**. In some embodiments, ILD layer **118** can include a dielectric material deposited using a deposition method suitable for flowable dielectric materials. For example, flowable silicon oxide can be deposited using flowable chemical vapor deposition (FCVD). In some embodiments, the dielectric

material can include silicon oxide.

[0039] In some embodiments, as shown in FIGS. 1 and 2, semiconductor device **100** can further include S/D contact structures **130**. In some embodiments, S/D contact structures **130** can be disposed on S/D structures **110**. In some embodiments, S/D contact structures **130** can include a silicide layer **130A** and a metal contact **130B**. In some embodiments, silicide layer **130A** can include metal silicide and can provide a lower resistance interface between metal contact **130B** and S/D structures **110**. Examples of metal used for forming the metal silicide include cobalt, titanium, and nickel. In some embodiments, metal contact **130B** can include conductive materials, such as tungsten, aluminum, and cobalt. In some embodiments, as shown in FIGS. 1 and 2, S/D contact structures **130** can extend through ILD layer **118** and into S/D structures **110**. In some embodiments, top surfaces of gate structures **112**, gate spacers **114**, ESL **116**, ILD layer **118**, and S/D contact structures **130** can be coplanar. In some embodiments, semiconductor device **100** can further include metal lines, metal vias, interconnects, and additional ILD layers.

[0040] FIG. 3 is a flow diagram of a method **300** for fabricating semiconductor device **100** having a semiconductor seed layer on S/D dielectric structures, in accordance with some embodiments. Method **300** may not be limited to nanostructure transistor devices and can be applicable to other devices that would benefit from the semiconductor seed layer on S/D dielectric structures.

Additional fabrication operations may be performed between various operations of method **300** and may be omitted merely for clarity and ease of description. Additional processes can be provided before, during, and/or after method **300**; one or more of these additional processes are briefly described herein. Moreover, not all operations may be needed to perform the disclosure provided herein. Additionally, some of the operations may be performed simultaneously or in a different order than shown in FIG. 3. In some embodiments, one or more other operations may be performed in addition to or in place of the presently-described operations.

[0041] For illustrative purposes, the operations illustrated in FIG. 3 will be described with reference to the example fabrication process for fabricating semiconductor device **100** as illustrated in FIGS. 4-12. FIGS. 4-12 illustrate partial cross-sectional views of semiconductor device **100** along line A-A as shown in FIG. 1 at various stages of its fabrication, in accordance with some embodiments. Elements in FIGS. 4-12 with the same annotations as elements in FIGS. 1 and 2 are described above.

[0042] In referring to FIG. 3, method **300** begins with operation **310** and the process of forming, on a substrate, a channel structure stacked on a fin structure. For example, as shown in FIG. 4, nanostructures **122** and nanostructures **420-1**, **420-2**, and **420-3** (collectively referred to as “nanostructures **420**”) stacked on fin structures **108** can be formed on substrate **104**. In some embodiments, nanostructures **122** and **420** can be stacked in an alternate configuration. In some embodiments, nanostructures **122** and **420** can be epitaxially grown on substrate **104** and subsequently patterned to form nanostructures **122** and **420** stacked on fin structures **108**. In some embodiments, nanostructures **122** and **420** can be in the form of nanosheets, nanowires, or nanoribbons. In some embodiments, nanostructures **122** and **420** can include semiconductor materials similar to or different from substrate **104**. In some embodiments, fin structures **108** can include the same semiconductor material as substrate **104**. In some embodiments, nanostructures **122** and **420** can include different semiconductor materials. For example, nanostructures **122** can include silicon, and nanostructures **420** can include silicon germanium with a germanium atomic percentage from about 10% to about 40%.

[0043] Embodiments of fin structures **108** and nanostructures **122** and **420** disclosed herein may be patterned by any suitable method. For example, the fin structures and the nanostructures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can combine photolithography and self-aligned processes, forming patterns that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For

example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers can be formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers can then be used to pattern the fin structures and the nanostructures.

[0044] The formation of nanostructures **122** can be followed by the formation of STI regions **106** between adjacent stacks of nanostructures **122** and **420** and the formation of sacrificial gate structures **412** on nanostructures **122**, as shown in FIG. 4.

[0045] Referring to FIG. 3, in operation **320**, a recess is formed in the fin structure adjacent to the channel structure. For example, as shown in FIG. 4, nanostructures **122** and **420** and fin structures **108** can be recessed to form recess **108r** in fin structures **108** adjacent to nanostructures **122** and **420**. In some embodiments, recess **108r** can extend into fin structures **108** below nanostructures **122** and **420**. In some embodiments, as shown in FIG. 4, recess **108r** in fin structures **108** can have a concave profile and a depth **108d** along a Z-axis ranging from about 4 nm to about 12 nm. In some embodiments, depth **108d** can ensure complete removal of bottom nanostructures **420** and formation of S/D dielectric structures **111**, semiconductor seed layer **113**, and amorphous portion **110A** of S/D structures **110** below the bottom surface of bottom nanostructures **122-1**.

[0046] In some embodiments, the formation of recess **108r** can be followed by the formation of inner spacers **121**, as shown in FIG. 4. In some embodiments, the formation of inner spacers **121** can include the lateral recess of nanostructures **420**, the deposition of a spacer layer, and the trim of the spacer layer.

[0047] Referring to FIG. 3, in operation **330**, a dielectric material is deposited in the recess on the fin structure. For example, as shown in FIG. 5, dielectric material **511** can be deposited in recess **108r** on fin structures **108** and on sacrificial gate structures **412**. In some embodiments, dielectric material **511** can be conformally deposited on fin structures **108**, nanostructures **122**, inner spacers **121**, and sacrificial gate structures **412**. In some embodiments, dielectric material **511** can be deposited by furnace, plasma-assisted atomic layer deposition (plasma-assisted ALD), conformal chemical vapor deposition (conformal CVD), or other suitable deposition method. In some embodiments, dielectric material **511** can include AlO.sub.x, SiC.sub.x, SiN.sub.x, SiC.sub.xN.sub.1-x, SiO.sub.yC.sub.xN.sub.1-x-y, a low-k material, or a combination thereof. In some embodiments, dielectric material **511** can have a thickness **511t** ranging from about 3 nm to about 5 nm.

[0048] Referring to FIG. 3, in operation **340**, a non-crystalline semiconductor material is deposited on the dielectric material. For example, as shown in FIG. 6, non-crystalline semiconductor material **613** can be deposited on dielectric material **511**. In some embodiments, non-crystalline semiconductor material **613** can be directionally deposited on dielectric material **511** by furnace, chemical vapor deposition (CVD) with potential bias, or other suitable deposition methods. In some embodiments, non-crystalline semiconductor material **613** can be deposited using a precursor of silane. In some embodiments, non-crystalline semiconductor material **613** can include amorphous or poly-crystalline semiconductor materials. In some embodiments, non-crystalline semiconductor material **613** can include non-crystalline silicon, non-crystalline silicon germanium, or other suitable non-crystalline semiconductor materials. In some embodiments, as shown in FIG. 6, non-crystalline semiconductor material **613** may be mainly deposited on top surfaces of dielectric material **511** and minimally deposited on sidewall surfaces of dielectric material **511**.

[0049] In some embodiments, the deposition of non-crystalline semiconductor material **613** can be followed by etching of non-crystalline semiconductor material **613**. For example, as shown in FIG. 7, non-crystalline semiconductor material **613** on sidewall surfaces of dielectric material **511** and on top surfaces of dielectric material **511** above sacrificial gate structures **412** can be removed by an etching process. In some embodiments, the etching process can use etching gases, such as hydrogen plasma and chlorine plasma. In some embodiments, the etching gases may not be able to reach non-crystalline semiconductor material **613** in recess **108r** and the remaining non-crystalline

semiconductor material **613** in recess **108r** can form semiconductor seed layer **113**.

[0050] In some embodiments, the etching of non-crystalline semiconductor material **613** can be followed by treating a top portion of semiconductor seed layer **113**. For example, as shown in FIG. **8**, the top portion of semiconductor seed layer **113** can be oxidized with an oxygen-containing source or nitridized with a nitrogen-containing source. In some embodiments, the top portion of semiconductor seed layer **113** can be oxidized or nitridized by a plasma process, a thermal process, an implantation process, or other suitable processes. In some embodiments, after the oxidation or nitridation treatment, the top portion of semiconductor seed layer **113** and dielectric material **511** can have a substantially same thickness, a substantially same density, and a substantially same wet etching rate. In some embodiments, as shown in FIG. **8**, the top treated portion of semiconductor seed layer can have a thickness **113d** along a Z-axis ranging from about 3 nm to about 5 nm.

[0051] Referring to FIG. **3**, in operation **350**, a portion of the dielectric material is removed to form a dielectric structure. For example, as shown in FIGS. **9** and **10**, a portion of dielectric material **511** on sacrificial gate structures **412**, nanostructures **122**, and inner spacers **121** can be removed to form S/D dielectric structures **111** and expose top surfaces of semiconductor seed layer **113**. In some embodiments, the portion of dielectric materials **511** can be removed by an isotropic etching process. In some embodiments, the isotropic etching process can include nitrogen trifluoride (NF.sub.3) dry etch, phosphoric acid with diluted hydrogen fluoride (H.sub.3PO.sub.4/HF) wet etch, or a dry etch and wet etch integration. In some embodiments, when thickness **113d** is greater than thickness **511t**, S/D dielectric structures **111** can have a “U” shape and end portions of S/D dielectric structures **111** can be above a top surface of semiconductor seed layer **113** and cover inner spacers **121** below bottom nanostructures **122-1**, as shown in FIG. **9**. In some embodiments, when thickness **113d** is equal to or less than thickness **511t**, end portions of S/D dielectric structures **111** can be at the same level or below the top surface of semiconductor seed layer **113**, as shown in FIG. **10**.

[0052] In some embodiments, as shown in FIG. **9**, the end portions of S/D dielectric structures **111** can have sloped top surfaces with inner edges of the end portions lower than the outer edges of the end portions. In some embodiments, with end portions of S/D dielectric structures **111** covering inner spacers **121** as shown in FIG. **9**, the parasitic capacitance between S/D structures **110** and gate structures **112** can be reduced and device performance can be further improved.

[0053] Referring to FIG. **3**, in operation **360**, an epitaxial structure can be grown on a top surface of the non-crystalline semiconductor material. For example, as shown in FIGS. **11** and **12**, S/D structures **110** can be epitaxially grown on top surfaces of non-crystalline semiconductor seed layer **113** and dielectric structures **111** shown in FIGS. **9** and **10**, respectively. In some embodiments, as shown in FIGS. **11** and **12**, S/D structures **110** can include a non-crystalline portion **110A** grown on semiconductor seed layer **113** and S/D dielectric structures **111** and a crystalline portion **110B** grown on non-crystalline portion **110A**. In some embodiments, non-crystalline portion **110A** can include a non-crystalline semiconductor material, such as amorphous or poly-crystalline semiconductor material. In some embodiments, non-crystalline portion **110A** can be below the bottom surface of bottom nanostructures **122-1**. If non-crystalline portion **110A** is above the bottom surface of bottom nanostructures **122-1**, the size of crystalline portion **110B** may be reduced, the strain imparted on the channel regions of nanostructures **122** may be reduced, and the carrier mobility in the channel regions of nanostructures **122** may be decreased. Additionally, if non-crystalline portion **110A** is above the bottom surface of bottom nanostructures **122-1**, nanostructures **122** may be in contact with non-crystalline portion **110A**, device current between nanostructures **122** and S/D structures **110** can be reduced and device performance can be degraded.

[0054] In some embodiments, crystalline portion **110B** of S/D structures **110** can include one or more epitaxial layers, where each epitaxial layer can have different compositions. In some

embodiments, each of the one or more epitaxial layers can include Si and differ from each other based on, for example, doping concentration and/or epitaxial growth process conditions. In some embodiments, each of the one or more epitaxial layers can include silicon germanium and differ from each other based on, for example, doping concentration, epitaxial growth process conditions, and/or relative concentration of germanium with respect to silicon.

[0055] In some embodiments, the formation of S/D structures **110** can be followed by the deposition of ESL **116** and ILD layer **118**, the replacement of sacrificial gate structures **412** with metal gate structures **112**, and the formation of S/D contact structures **130** on S/D structures **110**, as shown in FIG. 2. In some embodiments, the replacement of sacrificial gate structures **412** can include removal of sacrificial gate structures **412**, removal of nanostructures **420**, and deposition of metal gate structures **112**. In some embodiments, after deposition of gate structures **112** and formation of S/D contact structures **130**, a CMP process can planarize top surfaces of gate structures **112**, gate spacers **114**, ESL **116**, and ILD layer **118**, as shown in FIG. 2.

[0056] Various embodiments in the present disclosure provide methods for forming semiconductor seed layer **113** on S/D dielectric structures **111** in semiconductor device **100**. In some embodiments, nanostructures **122** stacked on fin structures **108** can be formed on substrate **104**. Recess **108r** can be formed in fin structures **108** adjacent to nanostructures **122**. S/D dielectric structures **111** and semiconductor seed layer **113** can be formed in recess **108r**. S/D dielectric structures **111** and semiconductor seed layer **113** can be below the bottom surface of bottom nanostructures **122-1**. S/D structures **110** can be formed on top surfaces of S/D dielectric structures **111** and semiconductor seed layer **113**. S/D structures **110** can include non-crystalline portion **110A** and crystalline portion **110B**. Crystalline portion **110B** can be in contact with nanostructures **122** and non-crystalline portion **110A** can be below the bottom surface of bottom nanostructures **122-1**. With S/D dielectric structures **111**, semiconductor seed layer **113**, and non-crystalline portion **110A** of S/D structures **110** below the bottom surface of bottom nanostructures **122-1**, resistance between S/D structures **110** and nanostructures **122** can be reduced. The carrier mobility in S/D structures **110** can be increased. The strain on nanostructures **122** exerted by S/D structures **110** can be increased. Accordingly, the DC device performance of semiconductor device **100** can be improved.

[0057] In some embodiments, a semiconductor structure includes a channel structure on a substrate, a gate structure wrapped around the channel structure, an inner spacer adjacent to end portions of the gate structure and the channel structure, a dielectric structure on the substrate and adjacent to the inner spacer, a non-crystalline semiconductor material on the dielectric structure, and an epitaxial structure on top surfaces of the non-crystalline semiconductor material and the dielectric structure.

[0058] In some embodiments, a semiconductor device includes first and second channel structures stacked on a fin structure and a dielectric structure on the fin structure and between the first and second channel structures. The dielectric structure extends into the fin structure and is below the first and second channel structures. a non-crystalline semiconductor material on the dielectric structure, and a source/drain (S/D) structure on a top surface of the non-crystalline semiconductor material. The S/D structure is adjacent to the non-crystalline semiconductor material and the dielectric structure.

[0059] In some embodiments, a method includes forming, on a substrate, a channel structure stacked on a fin structure, forming a recess in the fin structure adjacent to the channel structure, depositing a dielectric material in the recess on the fin structure, depositing a non-crystalline semiconductor material on the dielectric material, removing a portion of the dielectric material to form a dielectric structure, and growing an epitaxial structure on top surfaces of the non-crystalline semiconductor material and the dielectric structure.

[0060] It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as

contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.

[0061] The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a channel structure on a substrate; a gate structure wrapped around the channel structure; an inner spacer adjacent to end portions of the gate structure and the channel structure; a dielectric structure on the substrate and adjacent to the inner spacer; a non-crystalline semiconductor material on the dielectric structure; and an epitaxial structure on top surfaces of the non-crystalline semiconductor material and the dielectric structure.
2. The semiconductor structure of claim 1, wherein a top surface of the dielectric structure is below a bottom surface of the channel structure.
3. The semiconductor structure of claim 1, wherein a ratio of a thickness of the dielectric structure to a thickness of the channel structure ranges from about 0.2 to about 0.8.
4. The semiconductor structure of claim 1, wherein the epitaxial structure comprises a non-crystalline portion on the non-crystalline semiconductor material and a crystalline portion on the non-crystalline portion.
5. The semiconductor structure of claim 4, wherein a ratio of a thickness of the non-crystalline portion to a thickness of the channel structure ranges from about 0.3 to about 0.7.
6. The semiconductor structure of claim 4, wherein a top surface of the non-crystalline portion is below a bottom surface of the channel structure.
7. The semiconductor structure of claim 1, wherein a distance between a bottom surface of the dielectric structure and a bottom surface of the gate structure ranges from about 4 nm to about 12 nm.
8. The semiconductor structure of claim 1, wherein the dielectric structure comprises aluminum oxide, silicon carbide, silicon carbonitride, silicon nitride, silicon oxycarbonitride, or a low-k dielectric material.
9. The semiconductor structure of claim 1, wherein a ratio of a thickness of the non-crystalline semiconductor material to a thickness of the channel structure ranges from about 0.1 to about 0.9.
10. A semiconductor device, comprising: first and second channel structures stacked on a fin structure; a dielectric structure on the fin structure and between the first and second channel structures, wherein the dielectric structure extends into the fin structure and is below the first and second channel structures; a non-crystalline semiconductor material on the dielectric structure; and a source/drain (S/D) structure on a top surface of the non-crystalline semiconductor material, wherein the S/D structure is adjacent to the non-crystalline semiconductor material and the dielectric structure.
11. The semiconductor device of claim 10, wherein a top surface of the dielectric structure is below bottom surfaces of the first and second channel structures.
12. The semiconductor device of claim 10, wherein a ratio of a thickness of the dielectric structure to a thickness of the channel structure ranges from about 0.2 to about 0.8.
13. The semiconductor device of claim 10, wherein the S/D structure comprises a non-crystalline portion on the non-crystalline semiconductor material and a crystalline portion on the non-

crystalline portion.

14. The semiconductor device of claim 13, wherein a ratio of a thickness of the non-crystalline portion to a thickness of the channel structure ranges from about 0.3 to about 0.7.

15. The semiconductor device of claim 13, wherein a top surface of the non-crystalline portion is below a bottom surface of the channel structure.

16. The semiconductor device of claim 10, wherein a distance between a bottom surface of the dielectric structure and a bottom surface of the gate structure ranges from about 4 nm to about 12 nm.

17. A method, comprising: forming, on a substrate, a channel structure stacked on a fin structure; forming a recess in the fin structure adjacent to the channel structure; depositing a dielectric material in the recess on the fin structure; depositing a non-crystalline semiconductor material on the dielectric material; removing a portion of the dielectric material to form a dielectric structure; and growing an epitaxial structure on top surfaces of the non-crystalline semiconductor material and the dielectric structure.

18. The method of claim 17, wherein depositing the dielectric material comprises conformally depositing the dielectric material on the channel structure and the fin structure.

19. The method of claim 17, wherein removing the portion of the dielectric material comprises: treating a top portion of the non-crystalline semiconductor material to form a dielectric layer; and etching the dielectric layer on the non-crystalline semiconductor material and the dielectric material on the channel structure to expose the top surface of the non-crystalline material.

20. The method of claim 17, wherein growing the epitaxial structure comprises: growing a non-crystalline portion on the top surface of the non-crystalline semiconductor material, wherein a top surface of the non-crystalline portion is below a bottom surface of the channel structure; and growing a crystalline portion on the non-crystalline portion.
